Lesson 8: research plans linking environment and history

Reading Environment and History through Tree Rings / 90 minutes / Draft / 30 October 2026

Separate three evidential steps

Observation describes inspected or measured features. Dating assigns years using crossdating and calendar anchoring. Interpretation concerns felling, construction, disturbance, climate or activity. Ring formation, felling, use and deposition are different events, potentially separated by reuse or repair.

Combine applications

Archaeology examines objects, buildings and human activity; ecology examines trees and stands; climatology estimates a defined regional and seasonal variable. Recording, dating, comparison and explicit limits are shared, but a result in one field does not automatically answer a different field's question.

Examine causes

Coincident changes motivate a hypothesis. Water, food production, exchange, land use and institutions can mediate connections between climate and society. Check region, season and resolution before combining evidence. A causal diagram organizes hypotheses; its arrows do not establish their truth.

Locate uncertainty

Distinguish measurement and dating, preservation and sampling, and environmental or historical interpretation. Secure ring dates can coexist with unknown outer losses or post-felling intervals. Present conclusions together with these limits.

Reading a case: wooden containers, slips and public data

Comparing date evidence

Maeda and colleagues (2024) compare tree-ring dates of Nara wooden containers with written dates on wooden slips. Interpret what each date represents, including the last preserved ring, the event named in writing, association, preserved outer wood and the stages of use and deposition.

Nabunken records

The public dataset offers a route between images, measurement paths, values, identifiers and context. Verify identifiers before assuming that a database item and a paper concern the same material. This lesson introduces the sources by links and summaries without reproducing external figures or photographs.

Adding ecological and climate evidence

Growth suppression can involve competition or insects. Climate estimates require their region, season, units, calibration, verification and response stability. Distinguish independent evidence from another processing of the same source.

- R52 Early spring temperatures from Hinoki cypress - Yonenobu & Eckstein (2006)
- R55 Tree-ring dates of excavated wooden containers - Maeda et al. (2024)
- R56 Nara archaeological tree-ring dataset - 奈良文化財研究所 / Nabunken (2024)

Exercise A: three timber dates

All examples are fictional. A record refers to the building in 1210; samples come from different parts of the present structure. Spend 8 minutes recording claims, limits and additional evidence.

ID	Last preserved ring	Bark	Part
A	1187	Absent	1
B	1199	Present	2
C	1213	Absent	3

1. Which event and interval can each sample constrain?

2. What can be said about its relation to the 1210 record?

3. How could repair, reuse, storage, source interpretation and dating error be compared?

4. Which positions, joints, working traces or outer-edge details need examination?

Without bark, the last ring is not automatically the felling year. With bark, inspect intra-ring position and dormancy before specifying season. Date groups need structural context to support construction or repair interpretations.

Exercise B: your research plan

8 minutes: draft 5, review 2, revise 1. Do not describe unperformed observations or analyses as completed results.

Item	What to write
Question and target	Place, period and competing explanations.
Materials	Location, specimens, preservation and replication.
Methods	Observation, measurement, dating, analysis and independent checks.
Comparison	Expected observations under an alternative explanation.
Limits and next evidence	Uncertainty, contingencies and evidence that changes the claim.

Review

Does the evidence address the question? Are comparisons included? Which observation could support, revise or reject the explanation? Does the plan explain a procedure rather than simply list methods? Individual learners can review their own draft after a break.

Figures and records

Preserve measurements, units, missing values, identifiers and processing settings. Label axes, periods, sources and observed versus synthetic material. Credit source data in redrawn figures. Check conditions for particular external or collaboratively supplied materials.

Report, terms and reflection

Report structure

The syllabus indicates 2–4 A4 pages and at least three references. Connect question, sources, methods, discussion, conclusion and limits. Label a proposal and its expectations clearly. This material adds no new grading weights or deadlines.

Terms for searching

Observation, dating, interpretation, alternative hypothesis, association, redeposition, reuse and reproducibility. Use the bilingual glossary to connect original papers, reports and datasets across languages.

Reflection

1. How does a last-preserved-ring date differ from an event date?

2. Can you offer two explanations for one observation and specify evidence to compare them?

3. Name one observation that could change your own conclusion.

State the supported scope of a conclusion and distinguish unresolved points. Move among specimens, data and literature as you develop the next question.

- R08 International Tree-Ring Data Bank - NOAA NCEI
- R30 Interpreting tree-ring dates - Historic England
- R55 Tree-ring dates of excavated wooden containers - Maeda et al. (2024)
- R56 Nara archaeological tree-ring dataset - 奈良文化財研究所 / Nabunken (2024)